

INDEX + MATCH in Excel

1. What is INDEX + MATCH?

INDEX and MATCH are two functions used together to look up a value in a table. MATCH finds the position of an item in a list, and INDEX returns the value at that position from another column. Together they solve the problem of finding related information, such as the sales amount for a product. A Data Analyst uses them because they are more flexible than VLOOKUP: they can return a value from a column to the left of the search column, and they do not break when columns are inserted.

2. Syntax

`=INDEX(return_range, MATCH(lookup_value, lookup_array, 0))`

Argument	Required?	What it accepts
return_range	Required	The column (or row) that holds the value you want back (for example C2:C7)
lookup_value	Required	The value to find: text in quotes, a number, or a cell reference (for example B8)
lookup_array	Required	The column (or row) to search in (for example B2:B7). It must be the same size as return_range.
match_type	Optional*	0 exact match, 1 approximate (data sorted ascending), -1 approximate (data sorted descending)

*Always type 0 for an exact match. If it is left out, MATCH uses 1 and can return wrong results. MATCH(lookup_value, lookup_array, match_type) returns a position number. INDEX(array, row_num) returns the value at that position.

3. Simple Example

Find the sales amount for the product "Monitor" (typed in B8):

	A	B	C
1	Product ID	Product Name	Sales Amount
2	P001	Laptop	1200
3	P002	Mouse	150
4	P003	Keyboard	300
5	P004	Monitor	800
6	P005	Headphones	250
7	P006	Webcam	180
8	Result	Monitor	800

Formula in C8: `=INDEX(C2:C7, MATCH(B8, B2:B7, 0))`

Step 1: MATCH finds "Monitor" in B2:B7. It is the 4th item, so MATCH returns 4. Step 2: INDEX returns the 4th value in C2:C7.

Result: 800

4. Another Example

Find the name for employee ID E103. The name is to the left of the ID:

	A	B
1	Name	Employee ID
2	Aisha	E101
3	Rahim	E102
4	Nabila	E103
5	Tanvir	E104
6	Nabila	E103

Type the ID in B6. Formula in A6: `=INDEX(A2:A5, MATCH(B6, B2:B5, 0))`

MATCH finds E103 in B2:B5, in position 3. INDEX returns the 3rd value in A2:A5. **Result: Nabila**. VLOOKUP cannot return a column to the left of the search column, but INDEX + MATCH can.

5. When to Use It

- Find a sales amount, price, or other detail using an ID or name
- Return a value from a column to the left of the search column
- Pull specific data into reports and dashboards
- Build lookups that keep working when columns are inserted or moved
- Replace VLOOKUP in large tables

6. Important Things to Remember

- MATCH finds the position of the value in its range, not the sheet row number. INDEX uses that position in its own range.
- Use 0 as the last MATCH argument for an exact match. This is what you need in most cases.
- return_range and lookup_array must be the same size and start on the same row.
- Match types 1 and -1 need sorted data (ascending for 1, descending for -1).
- If MATCH finds no match, the result is #N/A. Matching is not case-sensitive.
- Text typed in the formula goes in quotation marks, or use a cell reference instead.

7. Common Mistakes

Mistake: Leaving out the 0, so MATCH uses approximate matching and returns wrong values. **Fix:** Always write , 0) at the end of MATCH.

Mistake: Searching in the wrong column, such as looking for "Monitor" in the column of product IDs, which gives #N/A. **Fix:** lookup_array must be the column that contains the value you are looking for.

Mistake: Ranges that do not line up, such as C2:C7 with B3:B8. **Fix:** Select the same rows in both ranges.

Mistake: Getting #N/A for a value that looks correct, because of extra spaces or different formats. **Fix:** Clean the data (for example with TRIM) and check that both sides are text or both are numbers.

8. Data Analyst Tip

Put the lookup value in a cell and lock the ranges with \$, such as \$B\$2:\$B\$7, so you can copy the formula down. Wrap the formula in IFERROR, for example `=IFERROR(INDEX(...), "Not Found")`, so missing items show a clear message instead of #N/A.